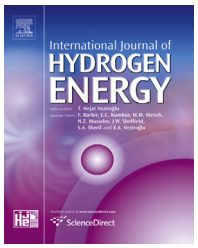
Available online at www.sciencedirect.com

ScienceDirect

journal homepage: www.elsevier.com/locate/he



Simulation of hydrogen and propylene coproduction in catalytic membrane reactor

E.V. Shelepova [a,*], A.A. Vedyagin [a,b], I.V. Mishakov [a,b], A.S. Noskov [a,b]

[a] Boreskov Institute of Catalysis, Siberian Branch, Russian Academy of Sciences, 5 pr. Lavrentieva, Novosibirsk 630090, Russia

[b] Novosibirsk State Technical University, 20 pr. Marksa, Novosibirsk 630073, Russia

ARTICLE INFO

Article history:
Received 6 June 2014
Received in revised form
24 July 2014
Accepted 2 September 2014
Available online 26 September 2014

Keywords:
Hydrogen production
Alkane dehydrogenation
Membrane reactor
Porous ceramic membrane
Pore size distribution

ABSTRACT

A two-dimensional non isothermal mathematical model has been developed to simulate the propane dehydrogenation in a catalytic membrane reactor with tube-and-shell configuration. The permeable inner tube consists of an inert large-pore support and thin microporous membrane layer. The membrane removes hydrogen from reaction zone shifting the reaction equilibrium towards products. Both pores diameter and membrane thickness were varied to obtain the optimal membrane characteristics in terms of hydrogen and propylene productivity, and hydrogen purity. The model correctly predicted the improved process parameters when the membrane thickness was 4 μm and the pore diameter was 0.4 nm.

Copyright © 2014, Hydrogen Energy Publications, LLC. Published by Elsevier Ltd. All rights reserved.

Introduction

Today many different techniques have been used for hydrogen production [1,2]. The following paths can be mentioned concerning hydrogen production: steam reforming of methane and natural gas [3–5], gasification of biomass [6,7] and etc. An alternative way is the dehydrogenation of alkanes with production of alkenes and hydrogen. To obtain the relatively pure hydrogen the separation procedure is required.

On the other hand, light olefins are important chemicals in organic synthesis because of their high chemical activity in certain reactions. Propylene is the one of the most valuable petrochemicals. The traditional processes for propylene production suffer from thermodynamic limitations, coke formation and require costly heat exchange at high operating temperatures because of the endothermicity of the reaction.

The traditional catalytic dehydrogenation is thermodynamically limited and requires very high temperatures (over 700 °C) to achieve a high enough conversion of propane. In addition, it has such disadvantages as the deactivation of the catalyst by coke formation, and the consequent need for continuous or periodic catalyst regeneration at frequent intervals throughout the process. While the propane conversion may be increased by operating at higher temperatures, that in its turn induces an increase in the rate of catalyst deactivation and a decrease in the reaction selectivity towards propylene.

In the last several decades, inorganic catalytic membrane reactors (CMR) have attracted extensive attention in the research community [8–10]. The use of a membrane allows one to remove selectively a product from reaction zone, shifting the reaction equilibrium towards the product side and simplifying subsequent product separations [11,12].

* *Corresponding author*. Tel.: +7 383 3269406; fax: +7 383 3306878.
E-mail address: shev@catalysis.ru (E.V. Shelepova).
http://dx.doi.org/10.1016/j.ijhydene.2014.09.004
0360-3199/Copyright © 2014, Hydrogen Energy Publications, LLC. Published by Elsevier Ltd. All rights reserved.

In this paper the propane dehydrogenation process was considered as an example. Propane dehydrogenation in the catalytic membrane reactor may be a potential method for increasing conversion while maintaining acceptable catalyst deactivation rate and reaction selectivity [13]. The essential benefit of integrating a selective membrane in the reactor is referring to higher propylene yield at lower temperatures in comparison with traditional catalytic dehydrogenation [14].

Thus, the application of the CMR permits one to combine the hydrogen and propylene production with separation in one reactor. In order to develop an effective membrane reactor, we have carried out the theoretical analysis and optimization of membrane characteristics in terms of their impact on process effectiveness.

Mathematical modeling

The membrane reactor scheme is represented by Fig. 1. It consists of two concentric tubes. The interior ceramic tube is filled with the fixed bed catalyst which is active in propane dehydrogenation reaction. A microporous oxide membrane covers the ceramic tube from the shell side. Depending on membrane pore size, hydrogen and other components of reaction mixture can permeate through the membrane to the shell compartment. Flow of inert sweep gas removes the permeated components from the shell side of reactor away. Selective hydrogen removal from the reaction zone shifts the reaction equilibrium towards the products increasing hydrogen and propylene yields. From other hand, relatively pure hydrogen can be obtained in the shell side of membrane reactor.

The developed mathematical model allows one to analyze the concentration profiles of all components in the reaction mixture as well as the propane conversion in terms of their distribution by the length of reactor. The equations of mass and energy balances are given below.

Mass balances

Tube side: $0 < r_1 < R_1$

$$\frac{\partial\left(u_l^t C_i^t\right)}{\partial l} = \varepsilon^t \frac{1}{r_1}\frac{\partial}{\partial r_1}\left(r_1 D_{e_i}^t \frac{\partial C_i^t}{\partial r_1}\right) + \rho_k^t(1-\varepsilon^t)\times\sum_{j=1}^{N_R}\gamma_{ij}w_j, \quad \forall i \tag{1}$$

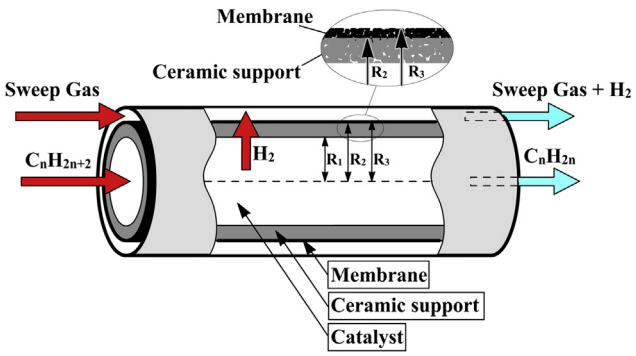


Fig. 1 – The scheme of the membrane reactor.

Boundary conditions:

$$l = 0:\ C_i^t = C_{in}^t;\ r_1 = 0:\ \frac{\partial C_i^t}{\partial r_1} = 0 \tag{2}$$

At the boundary ***tube/ceramic support:***

$$r_1 = R_1:\ C_i^t = C_i^c;\ D_{e_i}^t \varepsilon^t \frac{\partial C_i^t}{\partial r_1}\bigg|_{r_1=R_1} = D_{e_i}^c \varepsilon^c \frac{\partial C_i^c}{\partial r_2}\bigg|_{r_2=R_1}, \quad \forall i \tag{3}$$

Ceramic support: $R_1 < r_2 < R_2$

$$\frac{\varepsilon^c}{r_2}\frac{\partial}{\partial r_2}\left(r_2 D_{e_i}^c \frac{\partial C_i^c}{\partial r_2}\right) = 0, \quad \forall i \tag{4}$$

Boundary conditions:

At the boundary ***ceramic support/tube:*** the boundary conditions are the identical to those for *tube/ceramic support*.

At the boundary ***ceramic support/membrane:***

$$r_2 = R_2:\ C_i^c = C_i^m;\ D_{e_i}^c \varepsilon^c \frac{\partial C_i^c}{\partial r_2}\bigg|_{r_2=R_2} = D_{e_i}^m \varepsilon^m \frac{\partial C_i^m}{\partial r_3}\bigg|_{r_3=R_2}, \ \forall i \tag{5}$$

Membrane: $R_2 < r_3 < R_3$

$$\frac{\varepsilon^m}{r_3}\frac{\partial}{\partial r_3}\left(r_3 D_{e_i}^m \frac{\partial C_i^m}{\partial r_3}\right) = 0, \quad \forall i \tag{6}$$

At the boundary ***membrane/ceramic support:*** the boundary conditions are the identical to those for *ceramic support/membrane*.

At the boundary ***membrane/shell***:

$$r_3 = R_3:\ D_{e_i}^m \varepsilon^m \frac{\partial C_i^m}{\partial r_3}\bigg|_{r_3=R_3} = \beta\left(C_i^s - C_i^m\right), \ \forall i \tag{7}$$

Shell side:

$$\frac{\partial\left(u_l^s C_i^s\right)}{\partial l} = \beta\left(C_i^m - C_i^s\right), \ \forall i \tag{8}$$

Boundary conditions:

$$l = 0:\ C_i^s = 0,\ i = 1,\ N_t - 1 \tag{9}$$

Energy balance

Tube side: $0 < r_1 < R_1$

$$\rho_G^t c_p^t u_l^t \frac{\partial T^t}{\partial l} = \frac{1}{r_1}\frac{\partial}{\partial r_1}\left(\lambda_{ef}^t r_1 \frac{\partial T^t}{\partial r_1}\right) + \rho_k^t(1-\varepsilon^t)\times\sum_{j=1}^{N_R} w_j\left(-\Delta H_j\right) \tag{10}$$

Boundary conditions:

$$l = 0:\ T^t = T_{in}^t;\ r_1 = 0:\ \frac{\partial T^t}{\partial r_1} = 0 \tag{11}$$

At the boundary ***tube/ceramic support:***

$$r_1 = R_1:\ T^t = T^c,\ \lambda_{ef}^t \frac{\partial T^t}{\partial r_1}\bigg|_{r_1=R_1} = \lambda^c \frac{\partial T^c}{\partial r_2}\bigg|_{r_2=R_1} \tag{12}$$

Ceramic support: $R_1 < r_2 < R_2$

$$\frac{1}{r_2}\frac{\partial}{\partial r_2}\left(\lambda^c r_2 \frac{\partial T^c}{\partial r_2}\right) = 0 \quad (13)$$

Boundary conditions:

At the boundary ***ceramic support/tube:*** the boundary conditions are the identical to those applied for *tube/ceramic support*.

At the boundary ***ceramic support/membrane:***

$$r_2 = R_2:\ T^c = T^m,\ \lambda^c \frac{\partial T^c}{\partial r_2}\bigg|_{r_2=R_2} = \lambda^m \frac{\partial T^m}{\partial r_3}\bigg|_{r_3=R_2} \quad (14)$$

Membrane: $R_2 < r_3 < R_3$

$$\frac{1}{r_3}\frac{\partial}{\partial r_3}\left(\lambda^m r_3 \frac{\partial T^m}{\partial r_3}\right) = 0 \quad (15)$$

At the boundary ***membrane/ceramic support:*** the boundary conditions are the identical to those applied for *ceramic support/membrane*.

At the boundary ***membrane/shell***:

$$r_3 = R_3:\ \lambda^m \frac{\partial T^m}{\partial r_3}\bigg|_{r_3=R_3} = \alpha_1 (T^s - T^m) \quad (16)$$

Shell side:

$$\rho_G^s c_p^s u_l^s \frac{\partial T^s}{\partial l} = S_{sp1}\alpha_1 (T^m - T^s) + S_{sp2}\alpha_2 (T^w - T^s) \quad (17)$$

Boundary conditions:

$$l = 0:\ T^s = T_{in}^s \quad (18)$$

The model takes into account the alteration of thermophysical parameters of reaction mixture by the length and layer radius. Besides heat and mass distributions on reactor length and reactor radius, the model considers the volume change of gas mixtures due to reaction stoichiometry and components diffusion through the membrane. Change of mole amount is considered in the equation for calculation of axial velocity. The gas mixture velocities in the tube and shell sides were determined from the mass conservation equations. The corresponding equations were reported earlier for the case of ethylbenzene dehydrogenation in membrane reactor [15].

Parameters of mathematical model

The mass transfer coefficient was calculated from the following equation [16,17]:

$$\beta = \frac{\mathrm{Sh}\cdot D_i}{d_{eq}};\ \mathrm{Sh}_T = \mathrm{Sh}_\infty + f\left(\frac{d_i}{d_0}\right)\frac{0.19[\mathrm{Sc}(d_h/L)]^{0.8}}{1+0.117[\mathrm{Sc}(d_h/L)]^{0.467}};$$

$$\mathrm{Sh}_\infty = 3.66 + 1.2\left(\frac{d_i}{d_0}\right)^{-0.8};\ f\left(\frac{d_i}{d_0}\right) = 1 + 0.14\left(\frac{d_i}{d_0}\right)^{-0.5};$$

where d_0 – inner diameter of the outer tube, d_i – outer diameter of the inner tube, Sh – Sherwood number, Sc – Schmidt number (Prandtl diffusion number), $\mathrm{Sc} = \mu/(\rho_g D_m)$.

The effective coefficient of radial diffusion for ceramic support layer and membrane is described by the dependence: $D_e^{c,m} = ((1/D_m) + (1/D_{kn}))^{-1} \times perm \times b$. Knudsen diffusion coefficient: $D_{kn} = 2/3 r_{cap} u_i$, $u_i = \sqrt{8RT/\pi M_i/10^3}$. The effective coefficient of molecular diffusion is determined by Wilkes formula: $D_m^{c,m} = (1 - y_i)/\sum_{j=1,j\neq i}^{n}(y_j/D_{ij})$.

At modeling of mass transfer processes in microporous membrane the correction coefficient b was taken into account. The coefficient b takes into consideration the diffusion limitation which additionally appears at comparable values of pore diameter and molecule size.

Other dependences used for determination of mass and heat transfer coefficients were reported earlier [14]. The calculations of propane conversion and products selectivity were made under consideration of volume change of gas mixture [15].

The presented model is based on earlier developed and verified model for ethane dehydrogenation in catalytic membrane reactor with dense membrane [18].

It is well known that the diffusion processes in porous membrane consist of gas-phase Knudsen diffusion and adsorptive mass transfer which become dominated at low temperatures [19,20]. At high temperatures the adsorption processes are negligible and only gas-phase transfer is significant [20]. The process of alkanes dehydrogenation usually required temperatures above 400 °C when contribution of adsorption can be ignored. So, to describe the mass transfer processes inside the micropores of membrane only the Knudsen diffusion was taken into account. Steric factor was also taken into consideration, i.e. the diffusion coefficients were in a linear relationship with the pore diameter of membrane. As reported in literature the microporous membranes with pore diameter about 2 nm and more do not have appropriate perm selectivity for gas separation processes. These pores are too large to achieve essential separation factors [19]. As concluded the pore diameters should be less than 1 nm.

Reaction kinetics

Tube side

The reactions were considered in the tube compartment of the membrane reactor:

{1}. $C_3H_8 \leftrightarrow C_3H_6 + H_2$

{2}. $C_3H_8 \leftrightarrow C_2H_4 + CH_4$

{3}. $C_2H_4 + H_2 \leftrightarrow C_2H_6$

The kinetic parameters for these reactions on Pt–Sn–K/Al_2O_3 catalyst are given in Table 1.

The kinetics equations were obtained in a fluidized bed reactor (300 mm long tube) at atmospheric pressure: temperature, 500–540 °C; relative gas velocity, 1.2–4; minimum fluidization velocity at 500 °C, 0.96 $cm^3\ cm^{-2}\ s^{-1}$.

Numerical solution of the model equations

The mathematical model is presented by the system of partial derivatives equations. The system was transformed into the non-linear set of ordinary differential equations by the

Table 1 – The kinetic parameters for propane dehydrogenation process [21].

Reaction rate and rate constant equations	Reaction rate constant at T_0, mmol g^{-1} min^{-1} bar^{-1}	Activation energy, kJ mol^{-1}
$-w_{C_3H_8} = \frac{k_1(P_{C_3H_8} - (P_{C_3H_6}P_{H_2}/K_{eq}))}{1+(P_{C_3H_6}/K_{C_3H_6})}$ $k_1 = k_{01}\exp^{[-E_{a1}/R((1/T)-(1/T_0))]}$ $K_{C_3H_6} = K_0\exp^{[-\Delta H/R((1/T)-(1/T_0))]}$	$k_{01} = 0.5242$ $\Delta H = -85.817$ kJ mol^{-1} $K_0 = 3.46$	$E_{a1} = 34.57$
$-w_2 = k_2 P_{C_3H_8}$ $k_2 = k_{02}\exp^{[-E_{a2}/R((1/T)-(1/T_0))]}$	$k_{02} = 0.00465$	$E_{a2} = 137.31$
$-w_3 = k_3 P_{C_2H_4} P_{H_2}$ $k_3 = k_{03}\exp^{[-E_{a3}/R((1/T)-(1/T_0))]}$	$k_{03} = 0.000236$	$E_{a3} = 154.54$

method of lines. The received equations were treated numerically by the appropriate method for ODE solving. A second-order semi-implicit Rosenbrock algorithm with an automatic choice of the integration interval was used.

Simulation of the process

In this section, the variations of membrane pore size and thickness are investigated. The parameters values used for the simulation of the process in membrane reactor are listed in Table 2.

Fig. 2 illustrates the total amount of formed hydrogen as well as its distribution between tube and shell sides of membrane reactor. In this case, as an example, all calculations were done for the following characteristics of membrane: pore diameter, d_p = 0.3 nm (the minimal pore diameter in this study) and thickness of membrane layer, δ_m = 4 μm.

Curve 1 corresponds to total amount of hydrogen produced during the propane dehydrogenation. Each point of this curve was calculated as an amount of propylene formed according to reaction {1} with assumption that membrane is permeable for hydrogen only. The hydrogen concentration in tube side for the case of non-selective permeability of membrane is represented by curve 2. Hydrogen profile in the shell side of reactor was obtained as a remainder of curve 1 and curve 2.

It can be seen from the graph that the hydrogen concentration inside the tube grows up at the very beginning due to its formation and further falls down because of its removal through the microporous membrane to the shell side. Hydrogen profile in the shell side goes up monotonically, reflecting the amount of permeated hydrogen. Finally, 84.5% of produced hydrogen was removed from tube to shell side at the outlet of reactor.

Table 2 – The parameters of mathematical model used for the simulation of the propane dehydrogenation process in membrane reactor.

Parameter	Value	Parameter	Value
L, m	0.15	P^t, atm	1.5
r_1, m	0.39×10^{-2}	P^s, atm	1
d_r, m	0.2×10^{-1}	T^w, °C	500
δ_c, m	0.11×10^{-2}	$T^{t,s}_{in}$, °C	100
ε^t	0.5	G^t, ml/min	22
ε^c	0.28	G^s, ml/min	75
ε^m	0.14	$C^t_{C_3H_8,in}$, m.f.	0.1
d^c_p, m	1×10^{-6}	$C^t_{He,in}$, m.f.	0.9
d_k, m	0.15×10^{-2}	μ_g, kg m^{-1} s^{-1}	0.407×10^{-4}
ρ^t_k, g/m^3	0.2×10^7	$\lambda^{c,m}$, J m^{-1} s^{-1} K^{-1}	0.1

The pore diameter of membrane was varied while maintaining the membrane thickness similar. The resulted hydrogen profiles vs. reactor length in both tube and shell sides are presented in Fig. 3. As seen from the curves despite the minimal hydrogen concentration in tube side corresponds to membrane pore diameter of 0.5 nm, the maximum hydrogen concentration in shell side was obtained when the diameter of pore was 1 nm. An increase of $[H_2]_{tube}$ in the pore diameter range of 0.5–1 nm is related to non-selective permeation of all components of reaction mixture through membrane which minimize the order of equilibrium shift. It means that less amount of hydrogen was formed in this case. Percentage of hydrogen removal for this range is equal to 90%.

The calculated values of outlet hydrogen amount in shell side of reactor for different pores diameters are plotted in Fig. 4. The second curve on this figure represents a hydrogen purity which was obtained as molar ratio of permeated hydrogen to sum of all permeated components.

It is obvious that the hydrogen concentration in shell side grows up with increasing of pore diameter until d_p = 0.5 nm and then remains almost unchanged. On other hand the purity of hydrogen drastically drops down at 0.5 nm when the membrane becomes notably permeable for hydrocarbons. So, it can be concluded that effectiveness of the process along with hydrogen purity reach their maximum values in the case of pores with diameter of 0.4 nm. This result is in good

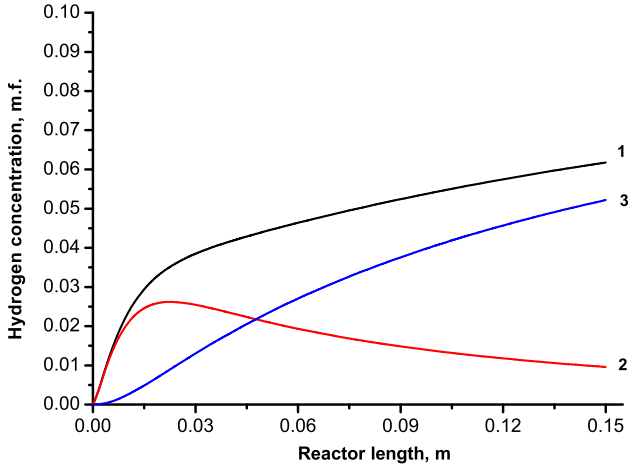


Fig. 2 – Hydrogen concentrations along the reactor length (d_p = 0.3 nm, δ_m = 4 μm).

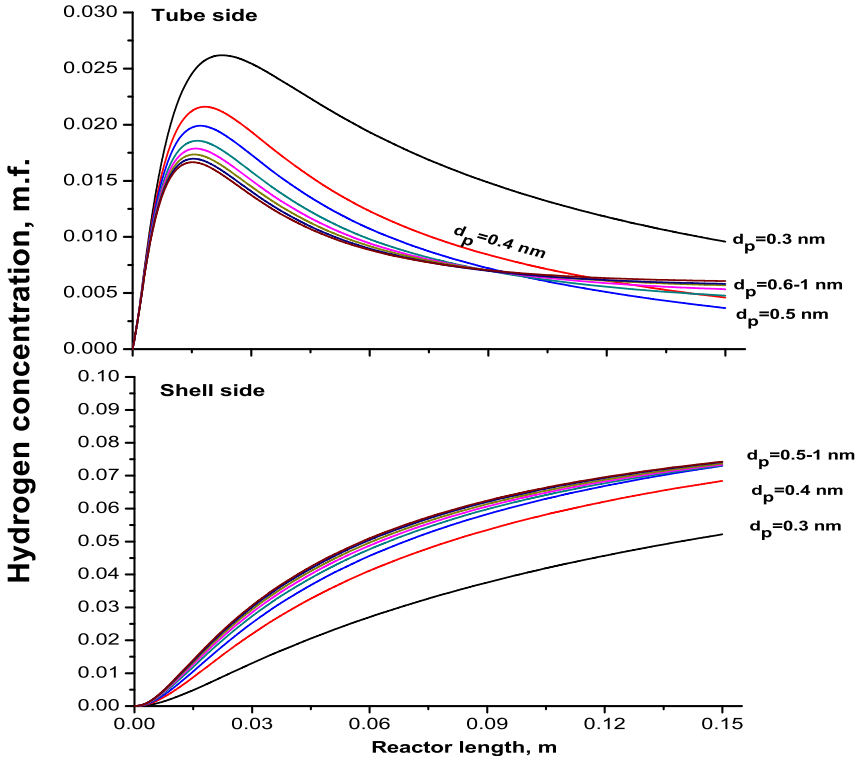


Fig. 3 – The hydrogen concentration profiles (mole fraction) vs. length of the reactor (meter) for various diameters of membrane pores ($\delta_m = 4$ μm).

agreement with literature data on gas separation by using the porous membranes [22].

The effect of membrane characteristics on the dehydrogenation process performance is of interest as well. The dependence of propane conversion, propylene selectivity and yield from the membrane pore diameter is presented in Fig. 5. The diameter was varied from 0.3 to 1.5 nm.

As can be seen from Fig. 5 all curves go through the maximum in the pore diameter range of 0.3–0.5 nm. The maximum value of propane conversion, 87%, is corresponded to $d_p = 0.5$ nm. The propylene yield decreased from 76 % to 41% with increasing of pore sizes from 0.4 to 1 nm. For the cases when pore size exceeded 1 nm, the membrane becomes fully non-selective in terms of permeation. The presence of membrane does not play a separation role anymore and the reaction equilibrium shift becomes insignificant. All values of propane conversion along with propylene selectivity and yield are equal to thermodynamically limited ones obtained earlier for the classic tube reactor without membrane. So, the

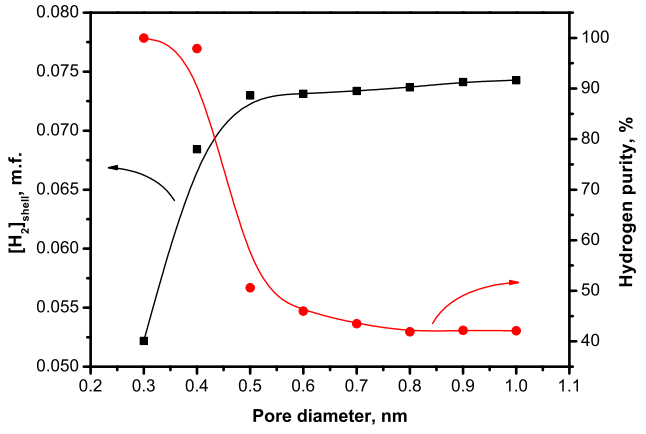


Fig. 4 – The hydrogen concentration (reactor outlet) and hydrogen purity profiles for various diameters of membrane pores ($\delta_m = 4$ μm).

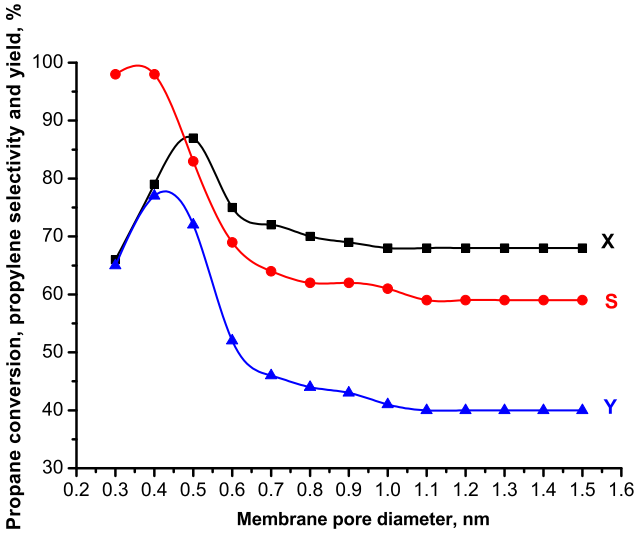


Fig. 5 – The propane conversion, propylene selectivity and yield (the reactor outlet) for various membrane pore diameters ($\delta_m = 4$ μm).

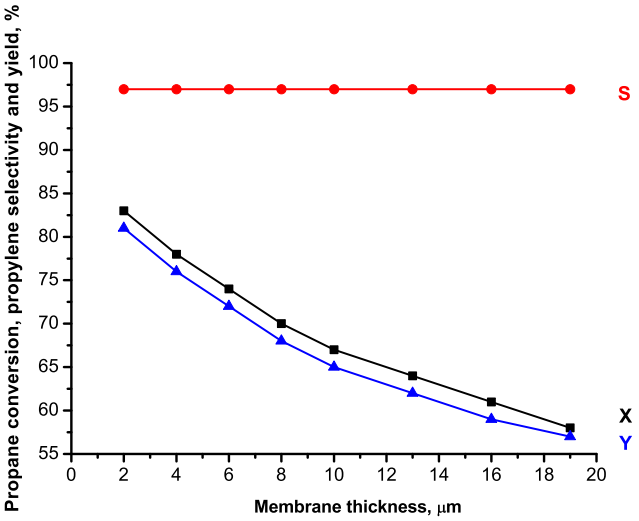


Fig. 6 – Propane conversion, propylene selectivity and yield on reactor outlet in tube side with varying of membrane thickness (d_p = 0.4 nm).

membrane pore diameter of 0.4 nm can be suggested as an optimum to obtain the maximum propylene yield.

In the final part of this research, in order to elucidate the influence of membrane thickness on the process performance, we have varied this characteristic in a wide range – from 2 to 19 microns. The pore diameter for these simulations was kept as 0.4 nm. As it is shown in Fig. 6, propane conversion has a tendency to decrease along with increase of membrane thickness. At the same time, the changes of membrane thickness does not influenced on selectivity of the process. It resulted in decreasing of propylene yield similarly to propane conversion. The maximum value of propylene yield (81%) was achieved at membrane thickness of 2 μm.

From other hand, the membrane thickness should be optimized from the point of view of hydrogen productivity and purity. The corresponding results of simulation are presented in Fig. 7. The enlargement of membrane thickness

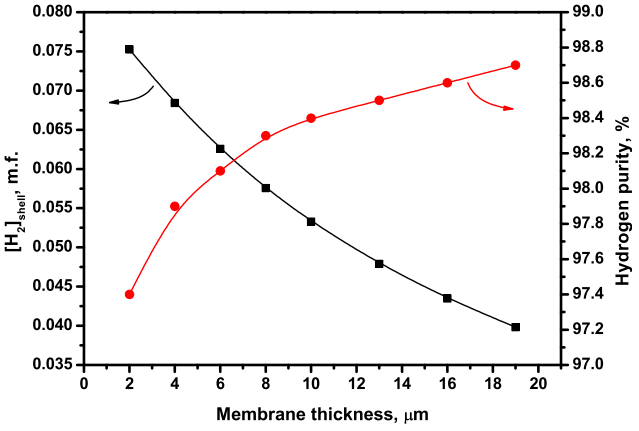


Fig. 7 – The hydrogen concentration on reactor outlet and hydrogen purity for various values of membrane thickness (d_p = 0.4 nm).

from 2 to 19 μm monotonically impairs hydrogen outlet concentration in 1.9 times. Observed improvement of hydrogen purity seems to be insignificant being varied between 97.4% and 98.7%. Here we can conclude that the optimal values of membrane thickness lay between 2 and 4 μm.

Conclusions

A two-dimensional mathematical model has been developed to simulate a process of dehydrogenation of propane in the membrane reactor.

The performed mathematical modeling was useful in defining the key characteristics of ceramic membrane including thickness and pore sizes at which the highest hydrogen and propylene productivity along with hydrogen purity can be reached.

The results of made simulations showed that the pore diameter of 0.4 nm is optimal in terms of hydrogen purity as well as of hydrogen and propylene productivity. The enlargement of membrane thickness leads to worse parameters; both propylene yields in tube side and hydrogen concentration in shell side drop down monotonically. Contrariwise it has no significant effect on hydrogen purity. The optimal values of membrane thickness were found to be limited between 2 and 4 μm.

REFERENCES

[1] Ahn SH, Choi I, Kwon OJ, Kim JJ. Hydrogen production through the fuel processing of liquefied natural gas with silicon-based micro-reactors. Chem Eng J 2014;247:9–15.
[2] Bang Y, Han SJ, Yoo J, Park S, Choi JH, Lee YJ, et al. Hydrogen production by steam reforming of liquefied natural gas (LNG) over mesoporous nickel–phosphorus–alumina aerogel catalyst. Int J Hydrogen Energy 2014;39(10):4909–16.
[3] Shu J, Grandjean BPA, Kaliaguine S. Methane steam reforming in asymmetric Pd- and Pd-Ag/porous SS membrane reactors. Appl Catal A General 1994;119:305–25.
[4] Nieva MA, Villaverde MM, Monzón A, Garetto TF, Marchi AJ. Steam-methane reforming at low temperature on nickel-based catalysts. Chem Eng J 2014;235:158–66.
[5] Olivieri A, Vegliò F. Process simulation of natural gas steam reforming: fuel distribution optimisation in the furnace. Fuel Process Technol 2008;89:622–32.
[6] Parthasarathy P, Narayanan KS. Hydrogen production from steam gasification of biomass: influence of process parameters on hydrogen yield – a review. Renew Energy 2014;66:570–9.
[7] Taylor AD, DiLeo GJ, Sunc K. Hydrogen production and performance of nickel based catalysts synthesized using supercritical fluids for the gasification of biomass. Appl Catal B Environ 2009;93:126–33.
[8] Gbenedio E, Wu Z, Hatim I, Kingsbury BFK, Li K. A multifunctional Pd/alumina hollow fibre membrane reactor for propane dehydrogenation. Catal Today 2010;156:93–9.
[9] Ziaka ZD, Minet RG, Tsotsis TT. A high temperature catalytic membrane reactor for propane dehydrogenation. J Memb Sci 1993;77(2–3):221–32.
[10] Bobrov VS, Digurov NG, Skudin VV. Propane dehydrogenation using catalytic membrane. J Memb Sci 2005;253(1–2):233–42.

[11] Gobina E, Hou K, Hughes R. Equilibrium-shift in alkane dehydrogenation using a high-temperature catalytic membrane reactor. Catal Today 1995;25:365–70.
[12] Tiemersma TP, Patil CS, van Sint Annaland M, Kuipers JAM. Modelling of packed bed membrane reactors for autothermal production of ultrapure hydrogen. Chem Eng Sci 2006;61:1602–16.
[13] Amer J, Ernst B, Burgard M. Permselectivity property of a based nickel membrane for its application to catalytic dehydrogenation of propane. Desalination 2006;200:700–1.
[14] Shelepova EV, Vedyagin AA, Mishakov IV, Noskov AS. Mathematical modeling of the propane dehydrogenation process in the catalytic membrane reactor. Chem Eng J 2011;176–177:151–7.
[15] Shelepova EV, Vedyagin AA, Mishakov IV, Noskov AS. Modeling of ethylbenzene dehydrogenation in catalytic membrane reactor with porous membrane. Catal Sustain Energy 2014;2:1–9.
[16] Heat exchanger design handbookvol. 2. Hemisphere; 1983.
[17] Lipnizki F, Field RW. Mass transfer performance for hollow fibre modules with shell-side axial feed flow: using an engineering approach to develop a framework. J Memb Sci 2001;193:195–208.
[18] Shelepova EV, Vedyagin AA, Noskov AS. Effect of catalytic combustion of hydrogen on dehydrogenation in a membrane reactor. II. Dehydrogenation of ethane. Verification of the mathematical model. Combust Explos Shock Waves 2013;49(2):125–32.
[19] Markovic A, Stoltenberg D, Enke D, Schlunder E-U, Seidel-Morgenstern A. Gas permeation through porous glass membranes part I: mesoporous glasses – effect of pore diameter and surface properties. J Memb Sci 2009;336:17–31.
[20] Markovic A, Stoltenberg D, Enke D, Schlunder E-U, Seidel-Morgenstern A. Gas permeation through porous glass membranes part II: transition regime between Knudsen and configurational diffusion. J Memb Sci 2009;336:32–41.
[21] Lobera MP, Tellez C, Herguido J, Menendes M. Transient kinetic modelling of propane dehydrogenation over a Pt–Sn–K/Al_2O_3 catalyst. Appl Catal A General 2008;349:156–64.
[22] Tennison SR. Microporous ceramic membranes for gas separation processes. Publ Rep 1998:68–74.

List of symbols

$C_i^{t,c,m,s}$: concentrations, kmol m^{-3}
$c_p^{t,s}$: heat capacity of gas mixture, kJ g^{-1} K^{-1}
$D_{e_i}^{t,c,m}$: effective coefficient of radial diffusion of component *i*, m^2 s^{-1}
D_{ij}: molecular diffusivity for component *i* in a binary mixture of *i* and *j*, m^2 s^{-1}
$D_m^{t,c,m}$: coefficient of molecular diffusion, m^2 s^{-1}
D_{kn}: Knudsen diffusion coefficient, m^2 s^{-1}
d_k: diameter of catalyst, m
d_r: diameter of membrane reactor, m
d_p^c: pore diameter of ceramic support, m
d_p: membrane pore diameter, nm
$G^{t,s}$: gas flow rate, ml min^{-1}
$-\Delta H_j$: heat effect of reaction *j*, kJ mol^{-1}
l: length of reactor, m
M_i: molecular weight of ith compound, g $mole^{-1}$
N_R: number of reactions within the tube side of reactor
$N_{t,s}$: number of components within the tube side/shell side of reactor
perm: permeability
P_0: pressure at normal conditions, atm
$P_i^{t,s,c,m}$: partial pressure of components, atm
$r_{1,2,3}$: radial coordinate into the tube side, in the ceramic support, in membrane, m
r_{cap}: capillary radius, m
R_r: reactor radius, m
R: universal gas constant, J mol^{-1} K^{-1}
S: selectivity, %
$S_{sp1,sp2}$: specific surface area, m^{-1}
T_0: temperature at normal conditions, K
$T^{t,s,c,m,w}$: temperature, K
u_i: average thermal velocity of molecule, cm s^{-1}
$u_1^{t,s}$: axial velocity, m s^{-1}
w_j: rate of reaction, kmol kg^{-1} s^{-1}
X: conversion, %
y_i: mole fraction of ith component
Nu_{we}: Nusselt number
Sc: Schmidt number (Prandtl diffusion number), $Sc = \mu/(\rho_g D_m)$
Sh: Sherwood number

Greek letters

$\alpha_{1,2}$: heat-transfer coefficient between the membrane and fixed bed catalyst (shell); between the exterior wall of reactor and fixed bed catalyst (shell), kJ m^{-2} s^{-1} K^{-1}
β: mass-transfer coefficient from membrane surface to shell side, m s^{-1}
$\delta_{c,m}$: ceramic support and membrane thickness, m
$\varepsilon^{t,c,m}$: porosity of catalyst layer (tube side); ceramic support and membrane
γ_{ij}: stoichiometric coefficient for i-component into j-reaction
λ_{ef}^t: effective coefficient of radial thermal conductivity, J m^{-1} s^{-1} K^{-1}
$\lambda^{c,m}$: thermal conductivity of the ceramic support, membrane, J m^{-1} s^{-1} K^{-1}
μ: dynamic viscosity of a gas mix, kg m^{-1} s^{-1}
$\rho_G^{t,s}$: gas density, kg m^{-3}
ρ_k^t: density of catalyst, kg m^{-3}

Indexes

c: ceramic support
in: inlet
m: membrane
s: shell side
t: tube side
W: wall of reactor
m.f.: mole fraction